SOFTWARE REQUIREMENTS SPECIFICATION

(IEEE 830-1998 Style)

SMART WASTE COLLECTION MANAGEMENT SYSTEM

Prepared for: Software Engineering Project

Document Version: 1.0

Date: August 2026

Table of Contents

1. Introduction

1.1 Purpose

1.2 Scope

1.3 Definitions, Acronyms, and Abbreviations

1.4 References

1.5 Overview

2. Overall Description

2.1 Product Perspective

2.2 Product Functions

2.3 User Characteristics

2.4 Constraints

2.5 Assumptions and Dependencies

2.6 Apportioning of Requirements

3. Specific Requirements

3.1 External Interface Requirements

3.2 Functional Requirements

3.3 Performance Requirements

3.4 Logical Database Requirements

3.5 Design Constraints

3.6 Software System Attributes

4. Appendices

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Smart Waste Collection Management System (SWCMS). The document is intended to provide a clear and complete specification for developers, project stakeholders, testers, administrators, and academic evaluators. It follows the prototype organization recommended by IEEE Std 830-1998, where requirements are stated in a form suitable for design and verification.

1.2 Scope

The Smart Waste Collection Management System is a software system designed to improve the planning, monitoring, and management of municipal or organizational waste collection activities. The system will maintain information about waste bins, their locations and fill status, collection workers, collection tasks, vehicles, citizen complaints, and collection reports.

The system aims to reduce overflowing bins, improve the efficiency of collection operations, provide better visibility to administrators, and allow citizens to report waste-related issues. The first version focuses on digital waste management operations and may support simulated or manually updated bin levels. Integration with physical IoT sensors and advanced route optimization may be implemented in future versions.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
SWCMS	Smart Waste Collection Management System
IoT	Internet of Things
GPS	Global Positioning System
Administrator	Authorized user responsible for managing system data and operations
Waste Collector	User responsible for carrying out assigned waste collection tasks
Citizen	User who can submit waste-related complaints or reports
Smart Bin	Waste bin whose fill status can be digitally monitored or updated

1.4 References

IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications. This document uses the IEEE 830 prototype SRS organization as the structural basis for this academic project specification.

1.5 Overview

Section 2 provides the overall context of the product, including its perspective, functions, users, constraints, and assumptions. Section 3 contains the specific, uniquely identified requirements of the system, including interface, functional, performance, database, and quality requirements. Section 4 contains supplementary information and future enhancements.

2. Overall Description

2.1 Product Perspective

SWCMS is a centralized information system that supports the coordination of waste management activities. It can operate as an independent web-based application connected to a database. Users interact with the system through role-based interfaces. In future deployments, the system may interface with GPS services, mapping services, mobile devices, and IoT-enabled waste bins.

The main modules are: User Management, Waste Bin Management, Collection Task Management, Complaint Management, Notification Management, and Reporting.

2.2 Product Functions

At a high level, the system will provide the following functions:

• Authenticate users and provide role-based access.

• Maintain records of waste bins and their locations.

• Record or receive waste-bin fill status information.

• Identify bins that require collection.

• Create and assign collection tasks to workers or vehicles.

• Allow collectors to update task progress and completion.

• Allow citizens to submit complaints regarding waste-related issues.

• Notify relevant users about important events.

• Generate operational reports and summaries.

2.3 User Characteristics

Administrators are expected to have basic computer knowledge and are responsible for managing bins, users, collection operations, complaints, and reports.

Waste Collectors should be able to use a simple web or mobile interface to view assigned tasks and update task status.

Citizens require only basic knowledge of using a web or mobile application to submit complaints and track their status.

2.4 Constraints

The system is subject to constraints such as internet availability, availability of compatible client devices, database server availability, and role-based security requirements. IoT and GPS integrations, if used, depend on external devices and services. The academic project implementation may use standard web technologies and a relational database.

2.5 Assumptions and Dependencies

It is assumed that users have access to an internet-connected device, administrators maintain correct system data, and collectors update collection status accurately. If IoT monitoring is implemented, it depends on the availability and reliability of sensors and network connectivity. Changes in these assumptions may affect system requirements.

2.6 Apportioning of Requirements

The following requirements may be delayed to future versions: real-time IoT sensor integration, AI-based prediction of waste generation, advanced GPS route optimization, push notifications, and a dedicated mobile application.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

UI-01: The system shall provide a login interface for authorized users.

UI-02: The system shall provide separate role-based dashboards for administrators, waste collectors, and citizens.

UI-03: The system shall provide forms for managing bins, collection tasks, and complaints.

UI-04: The system shall display clear status information for bins and collection tasks.

UI-05: The user interface shall provide validation messages when required input is missing or invalid.

3.1.2 Hardware Interfaces

HI-01: The system shall support standard desktop and mobile devices capable of running a modern web browser.

HI-02: If IoT sensors are integrated, the system shall accept fill-level data through a defined software communication interface.

3.1.3 Software Interfaces

SI-01: The system shall store persistent operational data in a relational database such as MySQL.

SI-02: The system may interface with mapping or GPS services when route or location functionality is enabled.

SI-03: The system shall expose or consume application interfaces using standard web communication protocols where external integrations are required.

3.1.4 Communication Interfaces

CI-01: Client-server communication shall use standard HTTP or HTTPS protocols.

CI-02: The system shall use secure communication for authentication and sensitive data in a production deployment.

3.2 Functional Requirements

3.2.1 User and Access Management

FR-01: The system shall allow authorized users to log in using valid credentials.

FR-02: The system shall identify the role of an authenticated user.

FR-03: The system shall restrict functions based on the authenticated user's role.

FR-04: The administrator shall be able to manage user accounts where user management is enabled.

3.2.2 Waste Bin Management

FR-05: The administrator shall be able to add a waste bin with a unique identifier and location details.

FR-06: The administrator shall be able to update waste-bin information.

FR-07: The administrator shall be able to mark inactive or remove obsolete waste bins.

FR-08: The system shall store the current status or fill level of each registered waste bin.

FR-09: The system shall allow the fill level to be updated manually or through an integrated external data source.

3.2.3 Collection Scheduling and Task Management

FR-10: The administrator shall be able to create a waste collection task for one or more waste bins.

FR-11: The system shall allow collection tasks to include priority, assigned collector, and scheduled date or time.

FR-12: The administrator shall be able to assign a collection task to an available waste collector.

FR-13: A waste collector shall be able to view assigned collection tasks.

FR-14: A waste collector shall be able to update task status as Pending, In Progress, or Completed.

FR-15: The system shall record the completion status and completion time of completed collection tasks.

3.2.4 Complaint Management

FR-16: A citizen shall be able to submit a waste-related complaint.

FR-17: A complaint shall contain an issue category and a description.

FR-18: The system shall assign a unique identifier to each submitted complaint.

FR-19: The administrator shall be able to view and update the status of complaints.

FR-20: The system shall maintain a history of complaint status information.

3.2.5 Notifications and Alerts

FR-21: The system shall generate an alert when a bin reaches a configured collection threshold.

FR-22: The system shall notify an assigned collector when a new task is assigned, subject to notification functionality being available.

FR-23: The system shall display relevant alerts and pending actions on the administrator dashboard.

3.2.6 Reporting

FR-24: The administrator shall be able to view summaries of waste bins and their current status.

FR-25: The administrator shall be able to generate reports of collection tasks by status and date range.

FR-26: The administrator shall be able to view summaries of citizen complaints and their resolution status.

FR-27: The system shall allow reports to be displayed on screen and may support export in a future implementation.

3.3 Performance Requirements

PR-01: Under normal academic project usage, the system should display standard dashboard pages within approximately three seconds, excluding external service delays.

PR-02: The system shall store valid user transactions without intentional loss of confirmed data.

PR-03: The system should support multiple concurrent users within the capacity of the deployed server environment.

3.4 Logical Database Requirements

The logical data repository shall maintain information required for the following entities:

DB-01: The system shall maintain data for Users: user identifier, role, authentication-related information, and profile details.

DB-01: The system shall maintain data for Waste Bins: bin identifier, location, status, fill level, and activity status.

DB-01: The system shall maintain data for Collection Tasks: task identifier, assigned collector, related bin or area, schedule, priority, and status.

DB-01: The system shall maintain data for Complaints: complaint identifier, reporting user, category, description, submission time, and resolution status.

DB-01: The system shall maintain data for Notifications: notification identifier, recipient, message, timestamp, and read status where applicable.

3.5 Design Constraints

DC-01: The system shall use a role-based access approach for administrative and operational functions.

DC-02: The implementation shall maintain unique identifiers for primary business entities.

DC-03: The system shall validate required input before storing data.

DC-04: The project implementation should use modular software components to support maintenance and future extension.

3.6 Software System Attributes

3.6.1 Security

NFR-01: The system shall require authentication before providing access to protected functions.

NFR-02: The system shall prevent users from accessing functions not authorized for their assigned role.

NFR-03: Sensitive authentication information shall not be displayed to unauthorized users.

3.6.2 Reliability

NFR-04: The system shall maintain consistent records when valid transactions are completed successfully.

NFR-05: The system shall provide an appropriate error message when an operation cannot be completed.

3.6.3 Usability

NFR-06: The system shall provide a user-friendly and understandable interface.

NFR-07: The system shall use clear labels and status indicators for major system functions.

3.6.4 Maintainability

NFR-08: The system should be organized into modular components to simplify modification and testing.

NFR-09: The system should maintain clear separation between user interface, business logic, and data access components.

3.6.5 Scalability

NFR-10: The system design should allow additional waste bins, users, and collection records to be added without requiring changes to existing user functions.

4. Appendices

Appendix A: Future Enhancements

Possible future enhancements include real-time IoT sensor integration, GPS tracking of collection vehicles, automated route optimization, AI-based prediction of bin filling patterns, recycling and waste segregation tracking, push notifications, and dedicated Android or iOS mobile applications.

Appendix B: Requirement Verification Approach

Each uniquely identified requirement in Section 3 should be verifiable through one or more of the following methods: demonstration, inspection, analysis, or testing. Functional requirements can primarily be verified through test cases; interface requirements through inspection and integration testing; and quality requirements through appropriate usability, security, reliability, and performance testing.